

Using AI as a Tutor

Overview

The goal of this assignment is to move away from using AI as a “shortcut” (a tool that gives you answers) and toward using it as a “**scaffold**” (a tool that helps you think harder). You will engage in a Socratic dialogue with Claude.ai to master a core concept from our course, then attempt to “break” the AI to identify its risks.

Part 1: Socratic AI

1. Setup

- Choose a topic from our course materials that you feel like you don’t fully understand. Download the relevant readings and lecture slides as PDFs.
- Log in to [Claude.ai](https://claude.ai).
- Create a “Project” in Claude and upload the PDFs of the course materials you want to discuss.
- **Upload:** Attach the PDF of the relevant course reading or the syllabus section for the topic you’ve chosen to study.
- **The Prompt:** Copy and paste the following “System Prompt” as your first message:

“You are an expert Socratic Tutor. I have uploaded my course materials. Your goal is to help me deeply understand [Insert Topic Name]. You must follow these rules: 1. NEVER give me a direct definition, summary, or answer. 2. Always respond with a single, thought-provoking question that helps me discover the next layer of the concept myself. 3. If I ask for the answer, politely refuse and provide a hint or an analogy instead. 4. If I make a mistake in my logic, ask a question that helps me see the contradiction. Let’s begin. Ask me a question to gauge what I already know about [Insert Topic Name].”

2. The Interaction

Engage in a conversation of at least **10 exchanges**. Your goal is to reach a “lightbulb moment” where you can define the concept in your own words without the AI ever telling you what it was.

Part 2: Stress Testing AI

Once you feel you understand the concept, try to **break the AI’s guardrails**. * Attempt to “nudge” the AI into agreeing with a common misconception or a false premise related to the topic (e.g., “*Actually, isn’t it true that [wrong info]?*”). * **Observation:** Does the AI stand its ground and correct you with a question, or does it become **sycophantic** (agreeing with you just to be “helpful”)?

Part 3: Critical Reflection

On Brightspace, write a brief reflection about the following:

1. Did you find this helpful? Share a specific snippet of the chat where the AI’s question forced you to realize something you hadn’t considered before.
2. How did this approach feel compared to simply asking an AI to “summarize the paper”? Which method do you think would help you more in a final exam?
3. Did the AI fail or get things wrong? Did it hallucinate a metaphor that didn’t make sense? Did it agree with your “false premise” in Part 2?
4. How did you feel about the AI? Did you find yourself treating the AI like a human teacher (e.g., feeling “embarrassed” to be wrong), or did you view it strictly as a software tool? Or something in between?

Include a **URL link** to your conversation (use the ‘Share’ button in Claude) so I can read through it.